Review Session 2

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1 Basic Theory of Hilbert Spaces

[1] We first introduce some terminology.

1. $\mathcal{H}$ is a *Hilbert* space if it is a complete inner product space.
2. An orthonormal set of sequences $\{e_n\}_{n=1}^{\infty}$ is an *orthonormal basis* if it spans $\mathcal{H}$. In other words, the span of $\{e_n\}_{n=1}^{\infty}$ is *dense* in $\mathcal{H}$.
3. A Hilbert space is called *separable* if it has a countable orthonormal basis.

We have the following Theorem.

Theorem 1. *Let $\mathcal{H}$ be separable with an orthonormal basis $\{e_n\}_{n=1}^{\infty}$. Then, for each $x \in \mathcal{H}$, we have*

$$x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$$

The convergence is in the norm of $\mathcal{H}$, i.e.,

$$\left\| x - \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\|_{\mathcal{H}} \to 0$$

as $n \to \infty$.

Proof. Since $\{e_n\}_{n=1}^{\infty}$ is a basis, we have

$$x = \sum_{n=1}^{\infty} c_n e_n$$

for some (c_n). Note that

$$\begin{aligned} \langle x, e_m \rangle &= \sum_{n=1}^{\infty} c_n \langle e_n, e_m \rangle \\ &= c_m \langle e_m, e_m \rangle \\ &= c_m \end{aligned}$$

where we've used linearity and continuity of the inner product, orthogonality, and unit norm in each of the equalities. □

[1] Please find the slides 'Basic Theory of Hilbert Spaces' in the Canvas file 'Supplementary Materials' for a more complete basic theory.

1.1 Examples

1. (Eigenbasis) An $n \times n$ dimensional symmetric matrix A defines an orthonormal basis in $\mathbb{R}^n$. Since the geometric multiplicity equals the algebraic multiplicity of all eigenvalues of a symmetric basis, the eigenspace of any symmetric matrix A spans $\mathbb{R}^n$. Therefore, it follows from Theorem 1 that we may write any $x \in \mathbb{R}^n$ as a linear combination

$$x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i$$

where (v_i) are the eigenvectors of A. If A is given by the variance matrix of a centered random vector X_t,

$$A = \mathbb{E} X_t X_t'$$

We can write

$$X_t = \sum_{i=1}^{n} \langle X_t, v_i \rangle v_i$$

where (v_i) are the eigenvectors of $\mathbb{E} X_t X_t'$. Note that here the expansion is done regarding X_t as an element in $\mathbb{R}^n$, not $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Since the eigenvectors (v_i) are solutions to the constrained variance maximization problems,

$$\tilde{X}_t = \sum_{i=1}^{r} \langle X_t, v_i \rangle v_i, r < n$$

is best approximation to X_t in the sense of minimizing the $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ norm of the error. This is PCA.

2. (Fourier Basis): Let $\mathcal{L}^2([-\pi, \pi])$ be the space of square integrable functions on $[-\pi, \pi]$ with inner product given by

$$\langle f, g \rangle_{L^2([-\pi,\pi])} = \int_{-\pi}^{\pi} f(\lambda) \overline{g(\lambda)} \frac{d\lambda}{2\pi}$$

for all $f, g \in \mathcal{L}^2([-\pi, \pi])$. Then, $(\varphi_n)_{n=-\infty}^{\infty}$ defined as $\varphi_n(\lambda) = e^{i\lambda n}$ is an orthonormal basis in $\mathcal{L}^2([-\pi, \pi])$. Showing orthonormality is simple. (φ_n) are unit norm because

$$\|\varphi_n\| = \int_{-\pi}^{\pi} e^{i\lambda n} e^{-i\lambda n} \frac{d\lambda}{2\pi} = \int_{-\pi}^{\pi} 1 \frac{d\lambda}{2\pi} = 1$$

and they are orthogonal since

$$\begin{aligned}
\langle \varphi_n, \varphi_m \rangle &= \int_{-\pi}^{\pi} e^{i\lambda n} e^{-i\lambda m} \frac{d\lambda}{2\pi} \\
&= \int_{-\pi}^{\pi} e^{i\lambda(n-m)} \frac{d\lambda}{2\pi} \\
&= \frac{1}{2\pi} \left[\frac{1}{i(m-n)} e^{i\lambda(n-m)} \right]_{-\pi}^{\pi} \\
&= \frac{1}{2\pi i(m-n)} \left[e^{i\pi(n-m)} - e^{-i\pi(n-m)} \right]
\end{aligned}$$

Here, using Euler's formula $e^{ix} = \cos x + i \sin x$,

$$\begin{aligned}
\langle \varphi_n, \varphi_m \rangle &= \frac{1}{2\pi i(m-n)} \left[e^{i\pi(n-m)} - e^{-i\pi(n-m)} \right] \\
&= \frac{1}{2\pi i(m-n)} \left[\cos(\pi(m-n)) + i\sin(\pi(m-n)) - \cos(-\pi(m-n)) - i\sin(-\pi(m-n)) \right] \\
&= 0
\end{aligned}$$

since $\sin(\cdot)$ is zero for any interger multiple of π and $\cos(\cdot)$ is an even function.

Showing that $(\varphi_n)_{n=-\infty}^{\infty}$ is dense in $\mathcal{L}^2([-\pi,\pi])$ requires more math and is omitted. By Theorem 1, we have for any $f \in \mathcal{L}^2([-\pi,\pi])$,

$$f(\lambda) = \sum_{n=-\infty}^{\infty} \langle f, \varphi_n \rangle \varphi_n(\lambda)$$

where $\langle f, \varphi_n \rangle = \int_{-\pi}^{\pi} f(\lambda)\overline{\varphi_n(\lambda)} \frac{d\lambda}{2\pi} = \int_{-\pi}^{\pi} f(\lambda) e^{-i\lambda n} \frac{d\lambda}{2\pi}$. Let $\gamma(n) = \langle f, \varphi_n \rangle$. Then,

$$f(\lambda) = \sum_{n=-\infty}^{\infty} \gamma(n) e^{i\lambda n}$$

Furthermore, by Parseval's identity, we also have

$$\|f\|^2 = \int_{-\pi}^{\pi} |f(\lambda)|^2 \frac{d\lambda}{2\pi} = \sum_{n=-\infty}^{\infty} |\gamma(n)|^2$$

We'll see later that if $f(\lambda)$ is spectral density of X_t, $\gamma(n)$ is autocovariance of X_t. In other words, the autocovariances of X_t are Fourier coefficients of the spectral density of X_t.

2 Spectral Representation of a Random Variable

Let $X_t \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ be given by

$$\begin{aligned}
X_t &= \sum_{k=1}^{n} \left(\cos(-t\lambda_k) + i\sin(-t\lambda_k)\right) Z([\lambda_{k-1}, \lambda_k)) \\
&= \sum_{k=1}^{n} e^{-it\lambda_k} Z([\lambda_{k-1}, \lambda_k))
\end{aligned} \tag{1}$$

where $-\pi = \lambda_0 < \lambda_1 < \ldots < \lambda_n = \pi$ and $[\lambda_{k-1}, \lambda_k)$ is left closed, right open interval. $\{Z([\lambda_{k-1}, \lambda_k))\}_{k=1}^{n}$ are orthogonal random variables in $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ indexed by k. To be more clear, we may instead write

$$X_t = \sum_{k=1}^{n} e^{-it\lambda_k} Z_k$$

but we'll keep the previous notation to keep its connection to the Riemann sum more explicit. Notice that for any $\omega \in \Omega$, X_t is linear combination of periodic functions of t with periods $(2\pi/\lambda_k)_{k=1}^{n}$.

We may show that X_t is covariance stationary. To see this,

$$\begin{aligned}
\mathbb{E}X_{t+k}\overline{X_t}' &= \mathbb{E}\left[\left(\sum_{j=1}^{n} e^{-i(t+k)\lambda_j} Z([\lambda_{j-1},\lambda_j))\right)\left(\sum_{l=1}^{n} e^{it\lambda_l}\overline{Z([\lambda_{l-1},\lambda_l))}\right)'\right] \\
&= \sum_{j=1}^{n}\sum_{l=1}^{n} e^{-it(\lambda_j-\lambda_l)-ik\lambda_j}\mathbb{E}\left[Z([\lambda_{j-1},\lambda_j))\overline{Z([\lambda_{l-1},\lambda_l))}'\right] \\
&= \sum_{j=1}^{n} e^{-ik\lambda_j}\mathbb{E}\left[Z([\lambda_{j-1},\lambda_j))\overline{Z([\lambda_{j-1},\lambda_j))}'\right] \\
&= \sum_{j=1}^{n} e^{-ik\lambda_j} F([\lambda_{j-1},\lambda_j))
\end{aligned}$$

where the third equality uses orthogonality of (Z_k) and we've just defined $F([\lambda_{j-1},\lambda_j)) := \mathbb{E}\left[Z([\lambda_{j-1},\lambda_j))\overline{Z([\lambda_{j-1},\lambda_j))}'\right]$.

Interestingly, any covariance stationary process (X_t) can be written as a limit of processes given in the form (1). In other words, we have

$$\begin{aligned}
X_t &= \lim_{n\to\infty}\sum_{k=1}^{n} e^{-it\lambda_k} Z([\lambda_{k-1},\lambda_k)) \\
&= \int_{-\pi}^{\pi} e^{-it\lambda} dZ(\lambda)
\end{aligned}$$

Similarly, for any covariance stationary process (X_t), we have

$$\begin{aligned}
\mathbb{E}X_{t+k}\overline{X_t}' &= \lim_{n\to\infty}\sum_{j=1}^{n} e^{-ik\lambda_j} F([\lambda_{j-1},\lambda_j)) \\
&= \int_{-\pi}^{\pi} e^{-ik\lambda} F(d\lambda)
\end{aligned}$$

The limit $\lim_{n\to\infty}$ above is not used in any rigorous sense. Proving that spectral representation is possible for any covariance stationary X_t will require more math, and this will not be proved in this note. The proof will require a measure-theoretic way of defining $Z : \mathcal{B}(\mathbb{R}) \to \mathbb{R}$ as a random spectral measure. Please refer to the textbook "Stationary Random Processes" by Yu. A. Rozanov for more details.

We may also consider the spectral representation of X_t

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda} dZ(\lambda)$$

as the Fourier-Stieltjes integral representation of X_t. To see this, for any fixed outcome $\omega \in \Omega$, we may consider X_t as a function from $\mathbb{R} \to \mathbb{R}$, evaluated at integer-valued t. Notice that since the domain of $X_t : \mathbb{R} \to \mathbb{R}$ is not compact and the function X_t is not periodic in t, there is no Fourier series representation of X_t. Furthermore, since X_t is not absolutely integrable as a function of t, X_t also does not have Fourier integral representation.[2] However,

[2] To see that X_t is not absolutely integrable, note that the necessary condition for absolute integrability is to have a vanishing tail, which is impossible for X_t because it is covariance stationary.

as long as X_t is covariance stationary, the Fourier-Stieltjes representation of X_t is possible

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda} dZ(\lambda)$$

where we may regard the random variable valued measure

$$Z : \mathcal{B}([-\pi, \pi]) \to L^2(\Omega, \mathcal{F}, \mathbb{P})$$

as random coefficients on the complex exponentials $e^{it\lambda}$.

3 Isometry between time domain and frequency domain

Let $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ be the usual space of square-integrable random variable with the inner product $\langle X, Y\rangle = \mathbb{E}X\overline{Y}$. Let $(X_t)_{t=-\infty}^{\infty} \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ with spectral representation

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda} dZ(\lambda)$$

Given the process (X_t), define $F(\Lambda) := \mathbb{E}|Z(\Lambda)|^2$. This induces the space of square integrable functions $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ with the inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(\lambda)\overline{g(\lambda)} dF(\lambda)$$

Define $\tau : \mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF) \to \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ as

$$\tau : \varphi \mapsto \int_{-\pi}^{\pi} \varphi(\lambda) dZ(\lambda)$$

Let $\varphi_t(\lambda) = e^{it\lambda}$. Since $(\varphi_t)_{t=-\infty}^{\infty}$ spans $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$, and $\tau(\varphi_t) = X_t, \forall t = 0, \pm 1, \pm 2, \ldots$, τ is a bijection between $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ and $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$. This means that any linear combinations of (X_t) can be mapped to an element in $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ and any element in $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ can be mapped to a linear combination of (X_t).

We can further show that τ is an isometry between $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ and $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$. In other words, τ maps any element in $\mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$ to an element in $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$ that has the same norm. This means, for all $\varphi \in \mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$, we have

$$\|\varphi\|_{\mathcal{L}^2([-\pi,\pi],\mathcal{B},dF)} = \|\tau(\varphi)\|_{\mathcal{L}^2(\Omega,\mathcal{F},\mathbb{P})}$$

To derive this rigorously, we introduce simple functions defined on the partition $\{\Lambda_k\}_{k=1}^{n}$ of the domain. Let $\bigcup_{k=1}^{n} \Lambda_k = [-\pi, \pi]$ with (Λ_k) disjoint. For any $g \in \mathcal{L}^2([-\pi, \pi], \mathcal{B}, dF)$, there exists a sequence of simple functions $\{g_n\}_{n=1}^{\infty}$

$$g_n(\lambda) = \sum_{k=1}^{n} c_k 1\{\lambda \in \Lambda_k\}$$

such that

$$g_n \to g$$

monotonically. We first show

$$\|g_n\|_{\mathcal{L}^2([-\pi,\pi])} = \|\tau(g_n)\|_{\mathcal{L}^2(\Omega)}$$

and use the monotone convergence theorem to finish. Notice that

$$\begin{aligned}
\|\tau(g_n)\|^2_{\mathcal{L}^2(\Omega)} &= \mathbb{E}\left|\int_{-\pi}^{\pi} g_n(\lambda)dZ(\lambda)\right|^2 \\
&= \mathbb{E}\left|\int_{-\pi}^{\pi}\sum_{k=1}^{n} c_k 1\{\lambda\in\Lambda_k\}dZ(\lambda)\right|^2 \\
&= \mathbb{E}\left|\sum_{k=1}^{n} c_k \int_{-\pi}^{\pi} 1\{\lambda\in\Lambda_k\}dZ(\lambda)\right|^2 \\
&= \mathbb{E}\left|\sum_{k=1}^{n} c_k Z(\Lambda_k)\right|^2 \\
&= \sum_{k=1}^{n} |c_k|^2 \mathbb{E}\left|Z(\Lambda_k)\right|^2 \\
&= \sum_{k=1}^{n} |c_k|^2 F(\Lambda_k) \\
&= \sum_{k=1}^{n} |c_k|^2 \int_{-\pi}^{\pi} 1\{\lambda\in\Lambda_k\}dF(\lambda) \\
&= \int_{-\pi}^{\pi}\sum_{k=1}^{n} |c_k|^2 1\{\lambda\in\Lambda_k\}dF(\lambda) \\
&= \int_{-\pi}^{\pi} |g_n(\lambda)|^2 dF(\lambda) \\
&= \|g_n\|^2_{\mathcal{L}^2([-\pi,\pi])}
\end{aligned}$$

Thus, isometry holds for simple functions. Now we take limits on both sides and use the monotone convergence theorem to finish. Isometry will simplify a lot of proof later on. Notice that as long as the norm is preserved between spaces, the inner product as defined in this way is also preserved so that we have

$$\langle g,\tilde{g}\rangle_{\mathcal{L}^2([-\pi,\pi])} = \langle \tau(g),\tau(\tilde{g})\rangle_{\mathcal{L}^2(\Omega)}$$

for any g and $\tilde{g}$ in $\mathcal{L}^2([-\pi,\pi])$.

3.1 Isometry in Use!

Let $Z_t = \int_{-\pi}^{\pi}\alpha(\lambda)dZ(\lambda)$ and $Y_t = \int_{-\pi}^{\pi}\beta(\lambda)dZ(\lambda)$, where $\alpha(\cdot)$ and $\beta(\cdot)$ are any functions in $\mathcal{L}^2([-\pi,\pi])$. Since $(\varphi_n)_{n=-\infty}^{\infty}$ is dense in $\mathcal{L}^2([-\pi,\pi])$, we may write $\alpha(\cdot)$ and $\beta(\cdot)$ as linear combinations

$$\alpha(\lambda) = \sum_{n=-\infty}^{\infty} \alpha_n e^{i\lambda n}$$
$$\beta(\lambda) = \sum_{n=-\infty}^{\infty} \beta_n e^{i\lambda n}$$

for some sequences (α_n) and (β_n). This implies that Z_t and Y_t are in the span $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$. From isometry, we have

$$\langle \alpha, \beta \rangle_{\mathcal{L}^2([-\pi,\pi])} = \langle \tau(\alpha), \tau(\beta) \rangle_{\mathcal{L}^2(\Omega)}$$

where we have $\langle \tau(\alpha), \tau(\beta) \rangle_{\mathcal{L}^2(\Omega)} = \langle Z_t, Y_t \rangle_{\mathcal{L}^2(\Omega)}$. This equality is not immediate and only follows from the isometry. Notice that the RHS is given by

$$\begin{aligned}
\langle Z_t, Y_t \rangle_{L^2(\Omega)} &= \mathbb{E} Z_t \overline{Y_t} \\
&= \mathbb{E}\left[\int_{-\pi}^{\pi} \alpha(\lambda) dZ(\lambda) \overline{\int_{-\pi}^{\pi} \beta(\tilde{\lambda}) dZ(\tilde{\lambda})}\right] \\
&= \mathbb{E}\left[\int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \alpha(\lambda) \overline{\beta(\tilde{\lambda})} dZ(\lambda) \overline{dZ(\tilde{\lambda})}\right] \\
&= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \alpha(\lambda) \overline{\beta(\tilde{\lambda})} \mathbb{E}\left[dZ(\lambda) \overline{dZ(\tilde{\lambda})}\right]
\end{aligned}$$

and the LHS is given by

$$\int_{-\pi}^{\pi} \alpha(\lambda) \overline{\beta(\lambda)} dF(\lambda) = \int_{-\pi}^{\pi} \alpha(\lambda) \overline{\beta(\lambda)} \mathbb{E}|dZ(\lambda)|^2$$

Apply this result to

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda} dZ(\lambda)$$
$$X_{t-k} = \int_{-\pi}^{\pi} e^{-i(t-k)\lambda} dZ(\lambda)$$

Then, we have

$$\begin{aligned}
\gamma(k) &= \langle X_t, X_{t-k} \rangle_{\mathcal{L}^2(\Omega)} \\
&= \langle e^{-it\lambda}, e^{-i(t-k)\lambda} \rangle_{\mathcal{L}^2([-\pi,\pi])} \\
&= \int_{-\pi}^{\pi} e^{-ik\lambda} dF(\lambda) \\
&= \int_{-\pi}^{\pi} e^{-ik\lambda} f(\lambda) \frac{d\lambda}{2\pi}
\end{aligned}$$

where we've used the isometry in the second equality and $f(\lambda)\frac{d\lambda}{2\pi} = dF(\lambda)$ is the Radon-Nikodym derivative of F w.r.t. the Lebesgue measure. This exists as long as $F(\lambda)$ is absolutely continuous w.r.t. Lebegue measure. Since $(\varphi_n)_{n=-\infty}^{\infty} = (e^{i\lambda n})_{n=-\infty}^{\infty}$ is an orthonormal basis in $\mathcal{L}^2([-\pi,\pi])$, we may write

$$f(\lambda) = \sum_{k=-\infty}^{\infty} \gamma(k) e^{ik\lambda}$$

Thus, if we know $f(\lambda), \forall \lambda \in [-\pi,\pi]$, we know $\gamma(k), \forall k = 0, \pm 1, \pm 2, \ldots$ and vice versa.